

Harry Lynch

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EDUCATION

Tufts University

Medford, MA

B.S. Computer Science, Minor in Mathematics | GPA: 3.8/4.0 (Major: 3.9)

Expected May 2027

- **CS Coursework:** Internet-Scale Distributed Systems, Parallel Computing, Networks, Database Systems, AI
- **Math Coursework:** Probability, Linear Algebra, Multivariable Calculus

EXPERIENCE

Bloomberg LP

New York, NY

Software Engineer Intern – Buy-Side Compliance, Enterprise Products

June 2026 – August 2026

- Owning the rewrite of the violation query/load path in the **C++23** backend of **VMGR**, Bloomberg's buy-side **compliance-violation monitoring** terminal function used by traders and auditors at major asset managers, hedge funds, and pension funds
- Pushing filtering of **100K+ row** violation datasets server-side and merging **real-time** violation rows from a message queue into the display, eliminating **minute-long load times** from bulk client-side loads
- Building a **multi-producer/multi-consumer queue** and **thread pool** moving database writes off the hot path

Crum and Forster

Morristown, NJ

Software Engineer / Data Science Intern

May 2025 – November 2025

- Cut state filing prep from a **1+ hour manual process** to **under 10 minutes** by building a Python tool that runs SQL queries in parallel, computes the actuarial figures, and auto-formats a regulator-ready exhibit
- Surfaced the strongest predictors of **claim frequency and severity** by building a Python (pandas) framework for univariate and bivariate analysis of **500K+ claim records**, informing variable selection for a pricing model
- Built a monthly Python **ETL pipeline** that loads **50K+ rows** of premium and claim data into a structured **SQL Server** schema, deduplicating records, validating data quality, and normalizing column names with regex

Launch (Startup)

Medford, MA

Full Stack Software Engineer Intern

January 2025 – May 2025

- Led a **5-intern backend team** designing a normalized **PostgreSQL** and **MongoDB** schema to model company, product, and user data
- Built a **RAG pipeline** (Ollama + scraped social data) powering an AI agent that generates company websites from existing profiles

PROJECTS

Shipping Container Inventory Platform | TypeScript, React, Node.js, PostgreSQL, AWS

May 2024 – Present

- **Sole engineer and operator** of a live, revenue-generating inventory and invoicing platform (**40K+ lines**, **275+ tests**) for a working shipping-container yard, tracking **248 containers** and **\$657K** in active inventory
- Built an **idempotent month-end billing engine** (cron) that derives storage charges from intake/checkout dates across three billing modes, generating **line-itemized, auditable** invoices safe to re-run without double-charging
- Automated deploys with a zero-touch **GitHub Actions** pipeline that builds Docker images and ships them to EC2, plus a **transactional migration runner** that backs up, applies, and validates **25 schema migrations**

Recursive Bayesian Radar-Trace Classifier | Python, NumPy, Pandas

May 2025

- Built a **recursive Bayesian estimator** from scratch in NumPy that classifies radar tracks as bird vs. aircraft from **600-sample velocity time series**, propagating a per-measurement posterior through a sticky-state **transition matrix** with NaN-robust updates
- Engineered a **rolling-volatility feature** with **Laplace-smoothed** empirical likelihoods from a 12,000-measurement corpus, fusing it with speed via **log-linear likelihood pooling** to lift held-out track accuracy from **9/10 to 10/10**

JumBuddy – Overall Winner & Education Track, JumboHack 2026

February 2026

- Architected the system and built the **capture pipeline** for a **VS Code extension**: keystroke-level edit tracking flushed to a **Flask** ingestion API, computing per-symbol metrics (thrash, churn, time-on-function) to help professors diagnose where students struggle

TECHNICAL SKILLS

Languages: C++, C, Python, SQL, TypeScript, JavaScript

Libraries & Frameworks: NumPy, pandas, scikit-learn, Seaborn, React, Node.js, Express, Flask, FastAPI

Tools: Linux, Git, Docker, GitHub Actions, AWS (EC2, S3, Textract), PostgreSQL, SQL Server, MongoDB

LEADERSHIP & ACTIVITIES

Tufts Varsity Football: Defensive Line

Aug 2023 – May 2024

Tufts Torn Ticket II: Stage Manager, Pit Guitarist, Actor

Aug 2023 – Present

Interests: Guitar, Welding, Theatre, Pickleball, Chess